



## KS3 Algebra

### Unit 1 Lesson

### 1: Simplifying Expressions

### Teaching Ideas

#### Learning Objective:

To simplify expressions by collecting like terms.

#### Success Criteria:

- To identify like terms in simple expressions.
- To simplify expressions containing more than one variable.
- To present a solution in its simplest terms.

#### Context

This is the first lesson of the new topic. It is expected that students will have little understanding of algebraic manipulation, but they should be confident in applying the order of operations.

#### Resources

Simplifying Expressions Powerpoint, Simplifying Expressions Matching Cards, Simplifying Expressions Triangular Dominoes, Collecting Terms Colour by Number and Simplifying Expressions Worksheet.

## Starter

Briefly explain some of the wider uses of algebra, and then present the problem of finding the cost of 3 pairs of identical shoes. Ask students how they would approach this.

Students are likely to answer that we just multiply the price of 1 pair by 3. However, explain that we might not know the cost and must therefore use algebra to solve the problem. Model the process of assigning a variable,  $f$ , to the cost of each pair and adding these together to get  $3f$ .

## Main Activities

### The Rules

Establish what is meant by 'variable' and 'like terms', working through the example to show these in use. Emphasise that a variable has an unknown value (if it had a known value, there would be no reason to use algebra!). It is also worth pointing out that a 'solution' in terms of simplifying an expression is not a single integer, but always contains at least one variable.

### Your Turn

There are 7 expressions for the pupils to simplify. Each question only contains a single variable, so the solution should only ever be written in terms of  $a$  or  $b$ . Answers are revealed on a click, giving you chance to clear up any misunderstanding. These questions are also included on the separate worksheet.

### The Rules

Now, introduce a problem involving two variables. Work through the example, modelling the process of collecting like terms to simplify the expression. It is important to note that different variables cannot be combined, because this requires multiplication (in this case, our solution is  $5a + 3b$  rather than  $8ab$ ).

### Your Turn

There are 3 expressions for the pupils to simplify. Each question contains more than one variable this time. Answers are revealed on a click, giving you chance to clear up any misunderstanding. Question 3 may need breaking down on the board, as it contains 3 variables. These questions are also included on the separate worksheet.

### Matching Cards

Hand out the **Simplifying Expressions Matching Cards**. Pupils should cut this into cards (or you may do so beforehand) and match the expressions with their simplified versions to create 8 pairs. Solutions are revealed on the next slide. Using the same cards, you can extend this into a memory game. All cards should be shuffled, then spread out and placed face down on the desk. Partners should take it in turns to flip over 2 cards. If the cards form a pair (as seen from the previous task), the pupil can claim this pair; if the cards do not form a pair, they should be flipped back over. The pupil who claims the most pairs wins.

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## Plenary

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Show students the 2D shapes. Each side of each shape has an algebraic value assigned to it. Pupils should use these to formulate an expression for finding the shapes' perimeters. This should then be expressed in its simplest terms. This is a good opportunity to iron out any misconceptions about adding unlike terms and confusing negatives with positives. Equally, you may indicate that our knowledge of shape properties should make some of these easier to solve.

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## Extra Practice

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If you would like to give your students more opportunities to practice simplifying expressions, you might consider using the included **Simplifying Expressions Triangular Dominoes** and **Collecting Like Terms Colour By Number**.